

LESSON 8-2 • SESSION 1 • PRACTICE • FORM B

Addition Equations

Learning target – I can solve an addition equation by subtracting the same amount from both sides. · 6.EE.B.7

REMEMBER

- **Inverse operations** undo each other. Subtraction undoes addition.
- Subtract the same number from **both sides** to keep the equation balanced.
- **Check** by putting your answer into the original equation.

WORKED EXAMPLE

A team had some points, then scored 14 more and has 40. Solve $y + 14 = 40$.

Step 1 Subtract 14 from both sides: $y + 14 - 14 = 40 - 14$.

Step 2 Simplify: $y = 26$.

Step 3 Check: $26 + 14 = 40$.

Answer $y = 26$ points

WORDS inverse operation (*operación inversa*) · isolate (*aislar*) · one-step equation (*ecuación de un paso*)

PRACTICE

show your work · label units

1. Solve each equation.

a. $n + 27 = 64$ ____

b. $16 + p = 43$ ____

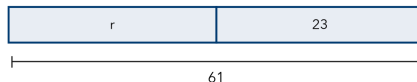
c. $x + 19 = 52$ ____

d. $y + 6.5 = 18$ ____

e. $m + 24 = 50$ ____

f. $w + 4.5 = 12$ ____

2. **Field Trip Seats.** A bus leaves school with r riders. At the next stop 23 more riders get on. Now there are 61 riders.



a. Write an addition equation. _____

b. Solve it. _____

c. Check your solution. _____

3. **Hiking Distance.** Mateo hiked 14.5 miles on Sunday. That was 4.5 miles more than he hiked on Saturday. Write and solve an equation for s , Saturday's miles.

Hint $\text{Saturday's miles} + 4.5 = \text{Sunday's miles}$.

Show your work.

Answer _____ miles

4. **Aquarium Fish.** A tank held f fish. The owner added 17 more. Then it held 63 fish. Solve $f + 17 = 63$.

Hint Subtract 17 from both sides.

Show your work.

f = _____

5. **Savings Jar.** Keisha had d dollars in a jar. She added \$18. Then the jar held \$81. Solve $d + 18 = 81$.

Show your work.

d = _____

- 6. Rainfall Total.** A rain gauge held r inches on Monday. On Tuesday it caught 1.8 more inches. The total was 6.3 inches. Write and solve an equation.

Hint Subtract 1.8 from both sides. Line up the decimal points.

Show your work.

Answer _____ inches

- 7. Book Pages.** Elijah read p pages last night. This morning he read 42 more pages, for 97 pages in all. Write and solve an equation. Check it.

Hint Undo + 42 on both sides.

Show your work.

Answer _____ pages

- 8. Savings Goal.** A student saved s dollars. After earning \$54 more, the total is \$135.

- Write an equation. _____
- Solve it. _____
- The student now wants \$200 in all. Write and solve an equation for e , the extra amount needed. _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Rashad's Work.** To solve $y + 34 = 50$, Rashad added 34 to both sides. He wrote $y = 84$. Explain his mistake. Then solve correctly.

TEST PRACTICE

choose the best answer

- 10.** Solve $8 + b = 23$.

A $b = 31$ **B** $b = 15$ **C** $b = 8$ **D** $b = 184$

- 11.** Solving an equation keeps it –

A larger **B** smaller **C** balanced **D** unequal

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

38 I	8.1 X	81 E	46 V	4.5 R	80 Z	55 S	63 E	84 Q	10 N
#2b		#3	#4	#5	#6	#7	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____